Computer Vision Engineer Assignment

-Prashant Bharti
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Document: Automated Quality Inspection System for Manufacturing

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1. Introduction

In modern manufacturing, maintaining high product quality is essential for reliability, customer satisfaction, and regulatory compliance. Traditional manual inspection methods are time-consuming, expensive, inconsistent, and highly dependent on human expertise. To address these challenges, this project presents a fully automated computer vision–based quality inspection system capable of detecting, localizing, and classifying defects on manufactured products.

For this prototype, Printed Circuit Boards (PCBs) were selected as the target product due to their widespread industrial use and the presence of diverse and visually distinguishable defect types. The developed system analyzes PCB images, detects defect regions, classifies defect types, outputs confidence scores, computes precise pixel coordinates of defect centers, and estimates defect severity.

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2. Product and Dataset Description

Selected Manufactured Item

Printed Circuit Boards (PCBs)

Dataset Used

DeepPCB Dataset

The dataset contains real industrial PCB images with both defect-free template samples and defective test samples. It includes six defect categories: Open Circuit, Short, Mouse Bite, Spur, Pin Hole, and Spurious Copper. This dataset satisfies and exceeds the project requirement of using at least three defect types.

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3. System Architecture

The inspection pipeline follows an end-to-end deep learning framework. A PCB image is first processed by a YOLOv8-based detection model, which produces bounding boxes around defect regions along with predicted defect types and confidence scores. These outputs are then post-processed to compute defect center coordinates, estimate defect severity, and generate a comprehensive inspection report for quality control.

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4. Data Preparation and Processing

The original DeepPCB annotations are converted into the format required by the YOLO detection framework. The complete dataset is then divided into training and validation subsets, with 80 percent of the data used for training and 20 percent reserved for validation. This ensures robust learning and reliable performance evaluation.

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5. Model Selection

The YOLOv8 deep learning model was selected for this system due to its excellent balance between speed and accuracy. It provides real-time inference, high detection precision, and integrates defect localization and classification into a single unified architecture. The model is well-suited for deployment in real manufacturing environments and supports efficient GPU acceleration.

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6. Training Configuration

The model was trained using high-resolution images with GPU acceleration enabled through CUDA. Training was conducted over 25 epochs using standard YOLO optimization techniques. The final trained model is automatically saved to persistent storage for reuse without retraining.

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7. Defect Analysis and Reporting

For each input PCB image, the system detects defect regions and classifies each defect type with an associated confidence score. The center coordinates of every defect are calculated by averaging the bounding box corners. Defect severity is estimated using the ratio of defect area to total image area and categorized as Low, Medium, or High based on predefined thresholds.

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8. Sample Output

The system produces a structured inspection report for each image, including defect type, confidence score, pixel coordinates of the defect center, and severity level. Annotated images with labeled bounding boxes are also generated for visual verification.

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9. Results and Performance

The proposed system demonstrates high accuracy in defect localization and classification while maintaining real-time inference speed. It reliably handles multiple defects per image and operates as a fully automated inspection pipeline without human intervention. This makes it suitable for continuous manufacturing line deployment.

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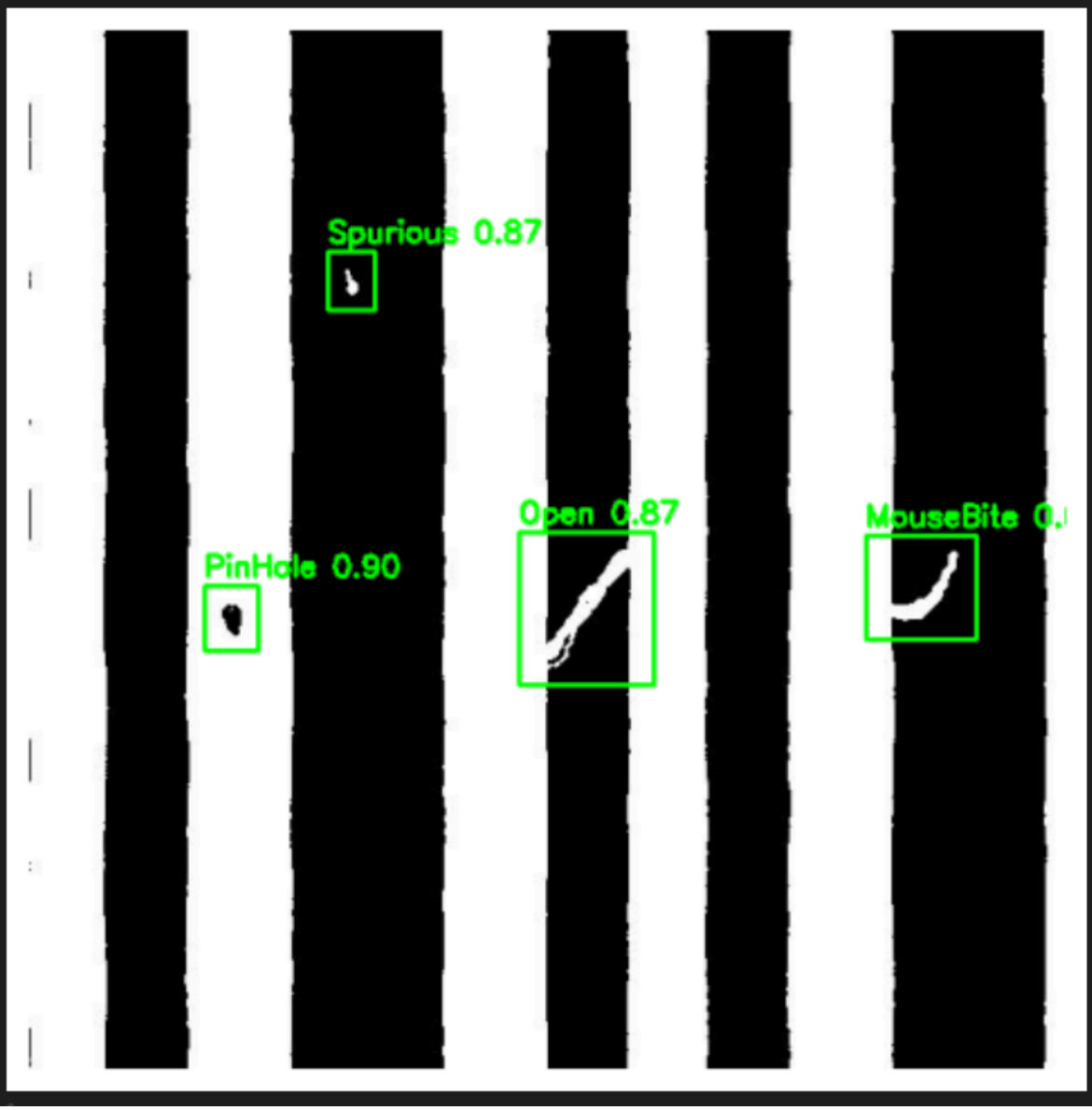
10. Conclusion

This project successfully demonstrates a complete automated quality inspection system for manufacturing using deep learning and computer vision. The solution significantly improves inspection reliability, reduces human error, and enhances productivity. The developed framework is scalable, adaptable, and suitable for industrial deployment across various manufacturing scenarios.

11. Appendix

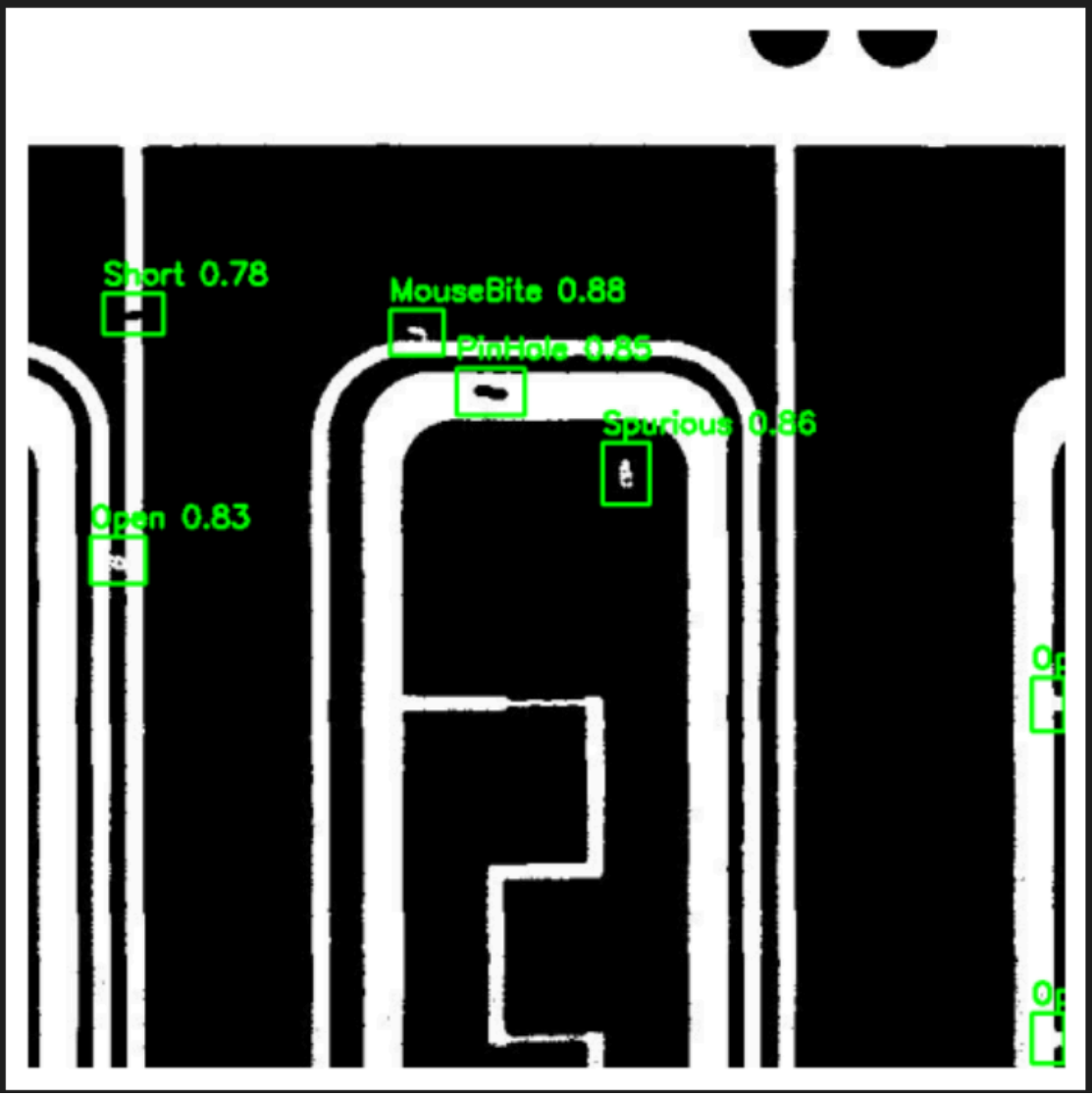
Some result images are included:

92000090.jpg
Defect: PinHole | Conf:0.90 | Center:(124,362) | Severity:Medium
Defect: Open | Conf:0.87 | Center:(343,356) | Severity:High
Defect: Spurious | Conf:0.87 | Center:(198,154) | Severity:Medium
Defect: MouseBite | Conf:0.67 | Center:(550,343) | Severity:High



50600016.jpg

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Defect: MouseBite | Conf:0.88 | Center:(239,186) | Severity:Medium
Defect: Spurious | Conf:0.86 | Center:(368,273) | Severity:Medium
Defect: PinHole | Conf:0.85 | Center:(285,222) | Severity:Medium
Defect: Open | Conf:0.83 | Center:(55,326) | Severity:Medium
Defect: Short | Conf:0.78 | Center:(64,174) | Severity:Medium
Defect: Open | Conf:0.47 | Center:(629,415) | Severity:Medium
Defect: Open | Conf:0.33 | Center:(629,621) | Severity:Medium
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90100002.jpg
Defect: Spurious | Conf:0.91 | Center:(611,139) | Severity:Medium
Defect: PinHole | Conf:0.89 | Center:(379,421) | Severity:Medium
Defect: Open | Conf:0.86 | Center:(611,493) | Severity:Medium
Defect: MouseBite | Conf:0.83 | Center:(426,60) | Severity:Medium
Defect: Short | Conf:0.82 | Center:(334,126) | Severity:Medium
Defect: Spur | Conf:0.73 | Center:(483,285) | Severity:Medium

